Library Management System

DBMS II CAT 2 Project Report

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Problem statement and purpose;

Libraries need an accurate way to record their books, members and borrowing activity. A paper register makes it difficult to know which books are available, which loans are overdue, and whether the same copy has been issued more than once. This project provides a web-based Library Management System backed by MySQL so that a librarian can complete these tasks from a browser.

System scope;

The system manages authors, categories, books, members, librarians and loans. Librarians sign in before accessing protected pages. They can add, view, edit and delete books and members, issue books, mark them as returned, search records and filter loan records by status. A dashboard gives a quick view of books, members, active loans and overdue loans.

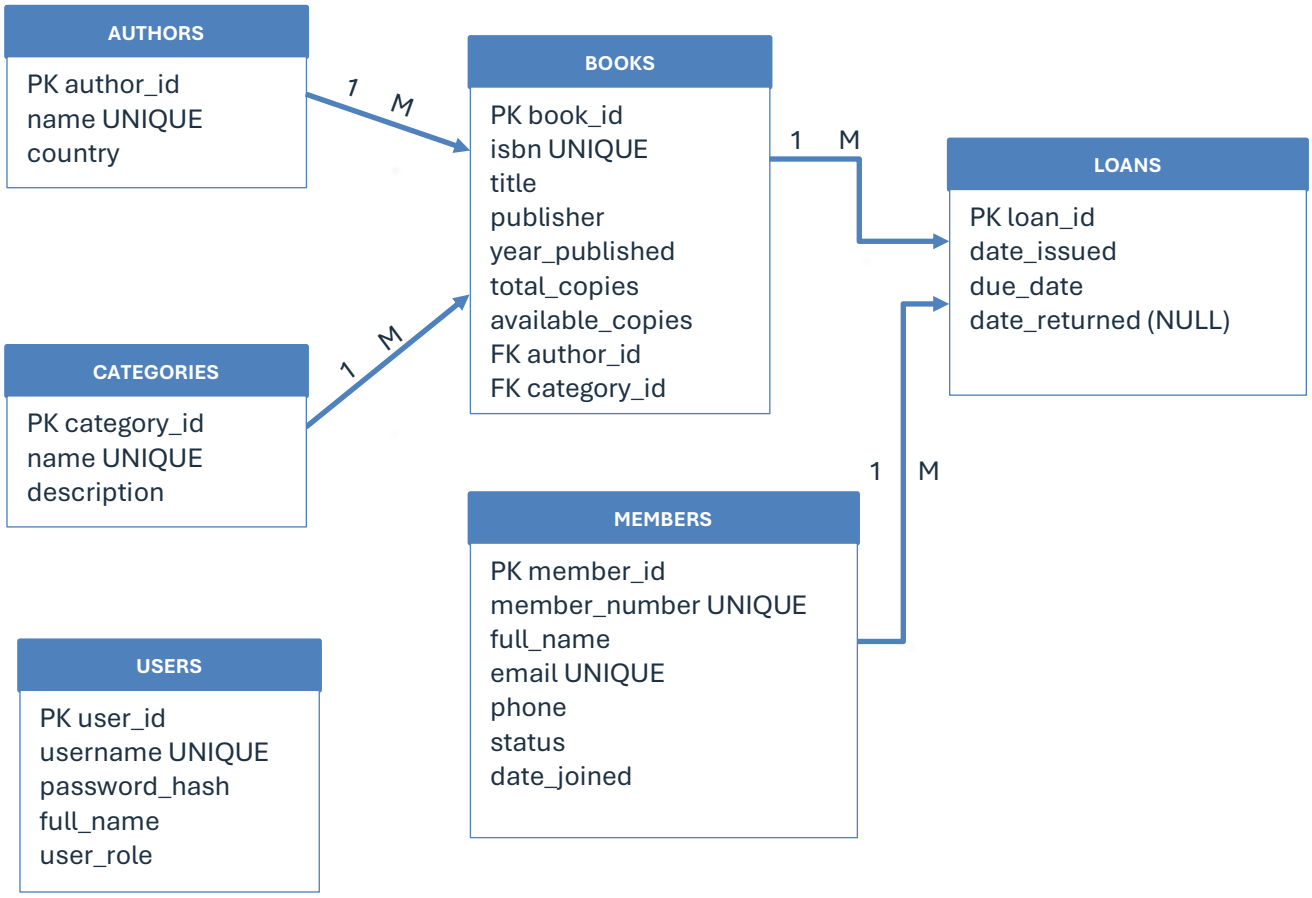
Development approach;

Our group designed the data structure first and then connected the PHP pages to it through PDO. We used AI as a learning and research aid to explore appropriate logic for the functionality required by the web application. In particular, it helped us expand our understanding of SQL relationships, transactions, prepared statements, PHP sessions and validation. The group reviewed the suggested approaches, implemented the project, and can explain the code and decisions made in it.

Database design and Entity Relationship Diagram

Library Management System Entity Relationship Diagram

PK = Primary Key FK = Foreign Key. One author/category can have many books; one member/book can have many loans.



Relationship is One to Many for example
AUTHORS → BOOKS;
CATEGORIES → BOOKS;
BOOKS → LOANS;
MEMBERS → LOANS.

Entity Relationship Diagram Description

The database has six related tables. Authors and categories describe a book, while members and books are connected through the loans table.

The loans table records the issue date, the due date and return date for each borrowing event.
Users are librarians who can log in to operate the system.

Primary keys identified are what uniquely identifies each record and foreign keys preserve the links between related record tables.

Normalization and data integrity

The design meets Third Normal Form. Each table describes one subject only. For example, author details are stored once in Authors rather than being repeated in every book record. Category details are also separated from books.
A loan contains only facts about one borrowing transaction and links to the correct member and book through foreign keys.

First, second and third normal form

Every table has a primary key, so each row is uniquely identifiable.

Columns store single values and there are no repeating groups, which satisfies First Normal Form.

Non key fields depend on the whole key in every table, satisfying Second Normal Form. Descriptive data such as author country and category description is stored in its own table rather than being dependent on another non key field, which satisfies Third Normal Form.

Constraints and security

The database uses NOT NULL, UNIQUE, DEFAULT, CHECK and FOREIGN KEY constraints where appropriate. ISBN, email, member number and username are unique.

PHP uses prepared statements for input, validates input on the server, escapes displayed values with htmlspecialchars, uses password_hash and password_verify for credentials, and protects private pages with PHP sessions.

Also, friendly messages are shown instead of raw database errors.

Core functions and business rules

Book and member management

Books and Members have complete browser based Create, Read, Update and Delete operations. Lists support search and filtering.

A confirmation prompt is shown before deletion, and the system explains why a record cannot be deleted when loan data depends on it.

Issuing a book

The application locks the selected member and book within a database transaction. It rejects an unknown selection, a suspended member, a book with no available copies, or a second active loan of the same book to the same member.

A successful issue creates the loan, sets the due date to fourteen days after issue, and reduces available copies by one.

Returning a book

A return is also handled in one transaction. The application finds and locks the active loan, records the return date, and increases the available copy count. If any part fails, the transaction is rolled back so that the loan and copy count cannot disagree.

Reports

The supplied SQL report file includes current loans with borrowers, overdue loans, and an aggregate count of loans by category. The dashboard also shows useful operational totals.

Testing and demonstration evidence

The following evidence should be captured after the project has been imported into XAMPP. Each screenshot should be saved in docs/screenshots and should show the browser address bar or page title where possible.

These captures are required because they demonstrate the working application instead of only the source code being tested.

1. The login page and a successful dashboard sign in
2. Book create, list or search, edit and delete confirmation
3. Member create, list or search, edit and delete confirmation
4. Book issue and a successful return
5. Rejected issue for an unavailable book, a suspended member and a duplicate active loan
6. A rejected deletion for a book or member with an unreturned loan
7. An overdue loan filter and responsive mobile width view

Conclusion

The Library Management System provides a normalized and secure solution for the main activities of a library. It combines a MySQL database, PHP business rules and a responsive browser interface.

The project is ready for demonstration after the included XAMPP setup steps are followed and the final evidence screenshots are captured.